Mathematics Gap Analysis
Shadow Rock Elementary

GEORGIA MILESTONES · GRADES 3–5 · SPRING RETEST 2026

PREPARED
May 2026

TOTAL STUDENTS
250 across 3 grades

⚠ Critical Finding: System-Wide Fraction & Operations Gap

Across all three grades, fractions and multiplicative reasoning show the highest rates of Below Target performance. The 5th grade cohort — which must retest — shows 67–86% Below Target across all six domains with zero students exceeding target in any area. Analysis indicates this is a

multi-year prerequisite gap

that began in 3rd grade and compounded through 4th into 5th. Re-teaching 5th grade standards alone will not close this gap.

5th Grade — Retest Cohort CRITICAL

78 STUDENTS RETESTING	**478** AVERAGE SCALE SCORE	**16** SPECIAL EDUCATION	**11** EL SERVICES

DOMAIN	BELOW TARGET RATE	BELOW / TOTAL	STATUS
Operations with Fractions NR · 5.NF		56 / 78 · 72%	CRITICAL
Numerical Patterns & Ordered Pairs PAR · 5.OA		54 / 78 · 69%	CRITICAL
Polygon Properties & Volume of Prisms GSR · 5.G / 5.MD		53 / 78 · 68%	CRITICAL

DOMAIN	BELOW TARGET RATE	BELOW / TOTAL	STATUS
Place Value and Decimals NR · 5.NBT		52 / 78 · 67%	CRITICAL
Multiplication, Division & Numerical Reasoning NR · 5.NBT		51 / 78 · 65%	CRITICAL
Measurement Conversions and Data Displays MDR · 5.MD		49 / 78 · 63%	HIGH

5th Grade · By Classroom

Goss, Tonya Michell

n=22 · Avg 455

Operations with Fractions	19/22 · 86%
Numerical Patterns & Ordered Pairs	19/22 · 86%
Polygon Properties & Volume	19/22 · 86%
Place Value and Decimals	17/22 · 77%
Multiplication, Division & NR	17/22 · 77%
Measurement Conversions	17/22 · 77%

Moon, Felicia Williams

n=12 · Avg 459

Multiplication, Division & NR	12/12 · 100%
Numerical Patterns & Ordered Pairs	12/12 · 100%
Operations with Fractions	11/12 · 92%
Place Value and Decimals	11/12 · 92%
Measurement Conversions	10/12 · 83%
Polygon Properties & Volume	9/12 · 75%

Johnson, Anthony

n=21 · Avg 491

Multiplication, Division & NR	14/21 · 67%
Numerical Patterns & Ordered Pairs	14/21 · 67%
Operations with Fractions	13/21 · 62%
Polygon Properties & Volume	13/21 · 62%
Place Value and Decimals	12/21 · 57%
Measurement Conversions	7/21 · 33%

Williams, Felycia Symone

n=23 · Avg 498

Measurement Conversions	15/23 · 65%
Operations with Fractions	13/23 · 57%
Place Value and Decimals	12/23 · 52%
Polygon Properties & Volume	12/23 · 52%
Numerical Patterns & Ordered Pairs	9/23 · 39%
Multiplication, Division & NR	8/23 · 35%

Prerequisite Gap Analysis · Fraction Strand

ROOT CAUSE

Why 5th Graders Cannot Do Fractions — The Gap Started in 3rd Grade

3RD GRADE Represent Fractions (3.NF) 38% Below Target	→	3RD GRADE Multiplication & Division within 100 (3.OA) 61% Below Target	→	4TH GRADE Add/Subtract Fractions (4.NF.B.3) 60% Below Target	→
4TH GRADE Decimals & Fractions (4.NF.C) 55% Below Target	→	5TH GRADE Operations with Fractions (5.NF) 72% Below Target			

Key Insight: Every node in the fraction prerequisite chain is broken. 5th graders who cannot add fractions with unlike denominators (5.NF.A.1) most likely never mastered equivalent fractions (4.NF.A.1) in 4th grade — and the 4th grade data confirms it. Those 4th graders likely never mastered fraction representation (3.NF) in 3rd grade. **Re-teaching 5th grade fraction standards without first assessing and filling the 3rd–4th grade prerequisite gaps will not produce passing scores on the retest.**

4th Grade · Prerequisite Pipeline

WARNING

88	499	64%	33%
STUDENTS	AVERAGE SCALE SCORE	BELOW TARGET · FRACTIONS	MET TARGET · ANGLES

DOMAIN	BELOW TARGET RATE	BELOW / TOTAL	5TH GRADE RISK
Addition and Subtraction of Fractions NR · 4.NF.B · feeds 5.NF directly		53 / 88 · 60%	DIRECT
Time and Data Displays MDR · 4.MD		50 / 88 · 57%	MODERATE

DOMAIN	BELOW TARGET RATE	BELOW / TOTAL	5TH GRADE RISK
Connections between Decimals and Fractions NR · 4.NF.C · feeds 5.NBT directly		48 / 88 · 55%	DIRECT
Number and Shapes and Patterns PAR · 4.OA		46 / 88 · 52%	MODERATE
Addition and Subtraction of Whole Numbers NR · 4.NBT		47 / 88 · 53%	MODERATE
Area, Perimeter and Polygons GSR · 4.MD / 4.G		49 / 88 · 56%	LOW
Angle Measurement GSR · 4.MD.C		40 / 88 · 45%	LOW

4th Grade · By Classroom · Fraction Pipeline

Brown, Christine Victoria
n=26 · Strongest Class

Add/Subtract Fractions	9/26 · 35%
Decimals & Fractions	4/26 · 15%

Dill, Zhane
n=25

Add/Subtract Fractions	16/25 · 64%
Decimals & Fractions	14/25 · 56%

Lewis, Cloreadia P
n=25 · Highest Risk

Add/Subtract Fractions	18/25 · 72%
Decimals & Fractions	15/25 · 60%

Simmons-Smith, Rose
n=12

Add/Subtract Fractions	10/12 · 83%
Decimals & Fractions	10/12 · 83%

3rd Grade · Foundation Assessment DEVELOPING

84 STUDENTS	496 AVERAGE SCALE SCORE	61% BELOW · MULT & DIVISION	38% BELOW · FRACTIONS

DOMAIN	BELOW TARGET RATE	BELOW / TOTAL	PIPELINE RISK
Length, Volume, Mass and Time MDR · 3.MD		53 / 83 · 64%	MODERATE
Multiplication and Division within 100 PAR · 3.OA · foundation for all fractions		51 / 83 · 61%	ROOT CAUSE
Attributes and Polygons GSR · 3.G		47 / 83 · 57%	LOW
Addition and Subtraction within 10,000 PAR · 3.NBT		44 / 83 · 53%	MODERATE
Area and Perimeter GSR · 3.MD.C		46 / 83 · 55%	LOW
Represent Fractions NR · 3.NF · prerequisite to all 4th-5th fraction work		32 / 83 · 38%	ROOT CAUSE

Recommended Intervention Priorities

ACTION REQUIRED

PRIORITY	ACTION	IMPACT ESTIMATE
1 — NOW	**Assess 5th graders on 3rd–4th grade fraction prerequisites before re-teaching any 5th grade standard.** Specifically: equivalent fractions (4.NF.A.1), adding fractions with like denominators (4.NF.B.3), and fraction representation (3.NF.A). Students who fail these assessments need prerequisite instruction, not 5th grade re-teaching.	Addresses 72% Below Target in 5.NF. Closes root cause, not symptom.

PRIORITY	ACTION	IMPACT ESTIMATE
2 — NOW	**Identify which 5th graders are in the 379–450 scale score band (24 students).** These students have significant gaps across all domains and likely need a fundamentally different instructional approach — small group intensive intervention — not whole-class re-teaching. Waiting for the retest without targeting this group will produce the same result.	24 students identified. Targeted intervention changes the retest trajectory.
3 — NEAR	**Flag the Dill, Lewis, and Simmons-Smith 4th grade classrooms for fraction pipeline monitoring.** These three classrooms produced 64–83% Below Target on 4.NF — the exact prerequisite the current 5th graders are missing. Without intervention, next year's 5th grade class enters with the same gap.	Prevents repeating this pattern in the 2026-27 5th grade cohort.
4 — NEAR	**Address 3rd grade Multiplication and Division within 100 (61% Below Target).** This is the computational foundation for all fraction understanding. Without multiplicative fluency, students cannot access fraction concepts meaningfully. This is the earliest intervention point in the pipeline.	Root cause intervention. Effects compound across 4th and 5th grade in 2026-28.
5 — MONITOR	**Note the Special Education (n=16) and EL Services (n=11) populations in the 5th grade retest cohort.** These students require accommodation review before the retest. Ensure IEP accommodations, EL language supports, and modified assessments are in place and documented.	Compliance and equity. 35% of retest cohort has identified learning needs.